Pledgement.

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on my honor. I have neither given nor received any unauthorized and/or inappropiate assistance for all sessions of this exam.

The work done on this exam is totally my own. I under stand by the school code, Violation of these Principles will lead to a zero grade and is subject to harsh discipls issuses."

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1. $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & 4 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

upper triangle $\begin{bmatrix} 1 & 0 & - & - \\ 0 & 1 & - & - \\ 0 & 0 & 1 & - \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$1\ \alpha\ -\gamma\ \alpha$

$0\ -3\alpha^2+5\ -3\alpha\gamma-9\gamma\ -3\alpha^2+1$

$0\ -2\alpha+\beta\ +2\gamma+3\ \alpha^2-2\alpha$

$0\ 14-\alpha\ \gamma-6\ 2-\alpha$

$\alpha \begin{bmatrix} 0 & 1 & 0 & +1 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha \\ 0 & 0 & 0 & 0 \end{bmatrix} + \beta \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & -9 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \end{bmatrix}$

they are the basis

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② $x - 5y = 0$

$$\begin{bmatrix} 1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

③ 1D is the minimum dim. inside 3D

④ $\begin{bmatrix} 1 & -5 & 0 \end{bmatrix}$

one Pivot $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 6$

x_1 $\quad x_1 = 2 \quad x_1 = 8$

$x_2 = 1 \quad x_2 = 0$

$x_3 = 0 \quad x_3 = 2$

yes they form a vector space because they are summable and scalable + 0 Vector

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③ given u is unit vector

and H is symmetric matrix such that $\rightarrow H = I - 2uu^T$

so $\|u\| = 1$

(a) Now.

$$H^2 = H \cdot H = (I - 2u.u^T)(I - 2.u.u^T)$$

$$= I^2 - 2I.u.u^T - 2I.u.u^T + 4u.u^T.u.u^T$$

$$= \underbrace{I}_{①}(I - 2u.u^T - 2u.u^T) + 4\,u.u^T.u.u^T$$

$$= I - 2u.u^T - 2u.u^T + 4u.u^T.u.u^T$$

$$= I - 2u.u^T - 2u.u^T + 4u.\|u\|^2.u^T$$

$\|u\|^2 = u^T u$

$\|u\| = \sqrt{u^T u}$

$\rightarrow \|u\| = 1$

So $H^2 = I - 4u.u^T + 4u.u^T$

$H^2 = I - 4(1) + 4(1)$

$H^2 = I$

(b)

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3.b $H \cdot H^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T)(I - 2(u^T)^T u^T)$ $\boxed{(AB)^T = B^T A^T}$

$= (I - 2uu^T)(I - 2uu^T)$

$= I - 2uu^T - 2uu^T + 4uu^T uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$ as we know $(\|u\| = 1)$ same as Part A

$= I - 4uu^T + 4uu^T$

$= I$

So $H^T \cdot H = I$

So H is an orthogonal Matrix

(3.c) $Hu = (I - 2uu^T) \cdot u$

$= u - 2u(u^T u)$ $\boxed{u^T u = 1}$

$= u - 2u$

$= -u \rightarrow (-1)u = -1$

So -1 is the eigen Value for H

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3(B) Given $H = (I - 2uu^T)$

As $H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2(u^T)^T u^T$

$= I - 2uu^T = H$

So H is a symmetric Matrix

and: $H = S \Lambda S^T$

$= S \Lambda S^{-1}$

Where S is an orthogonal Matrix

$(S^{-1})^{-1} \Lambda S^{-1}$

As $SS^T = I \rightarrow S^T = S^{-1}$

$P^{-1} \Lambda P$

So H is diagonalizable Matrix

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(4). because they are connected to each other

so we have continus system

$V(0) = 10$

$W(0) = 40$

$$\frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} V \\ W \end{bmatrix}$$

transtion matrix

$$\det\begin{bmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{bmatrix} = 0 \qquad \det(A-\lambda I)=0$$

$$2-3\lambda+\lambda^2-2=0$$

$$\lambda^2-3\lambda=0$$

$$\lambda(\lambda-3)=0$$

$$\lambda_1=0 \quad \lambda_2=3$$

$$X_1=\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow X_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$X_2=\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} x_3 \\ x_4 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow X_2=\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$U_K = \lambda_1^K \cdot S_1 \cdot C_1 + \lambda_2^K S_2 \cdot C_2$$

$$\begin{bmatrix} V_K \\ W_K \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot C_1 + 3^K \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix} \cdot C_2$$

at $t=0$

$$\begin{bmatrix} 40 \\ 10 \end{bmatrix} = \begin{bmatrix} -2C_2 \\ C_2 \end{bmatrix}$$

i couldn't find C_2 here so the correct transtion matrix is

to find the steady state

$K \rightarrow \infty$

$$\frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & \end{bmatrix}\begin{bmatrix} V \\ W \end{bmatrix}$$

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